Question ID: e4a588ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = \frac{1}{2}(x + 6)$. What is the value of $f(4)$?

Answer

A. 20

B. 12

C. 10

D. 5

Correct Answer: D

Rationale

Choice D is correct. It's given that the function f is defined by $f(x) = \frac{1}{2}(x + 6)$. Substituting 4 for x in the given function yields $f(4) = \frac{1}{2}(4 + 6)$, or $f(4) = 5$. Therefore, the value of $f(4)$ is 5.

Choice A is incorrect. This is the value of $2(4 + 6)$, not $\frac{1}{2}(4 + 6)$.

Choice B is incorrect. This is the value of $2 + (4 + 6)$, not $\frac{1}{2}(4 + 6)$.

Choice C is incorrect. This is the value of $4 + 6$, not $\frac{1}{2}(4 + 6)$.